

Robert Guy

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Research Overview

I study scheduling and resource arbitration in real-time distributed systems, with a focus on tail-tolerant performance under contention, uncertainty, and strict latency constraints. My research develops predictive and adaptive scheduling policies augmented with explicit runtime guard mechanisms (policy enforcement) to control extreme latency behavior.

Education

Georgia Institute of Technology M.S., Electrical & Computer Engineering August 2024 – May 2026

- Major GPA: 3.9/4.0

East Carolina University B.S., Computer Science January 2018 – May 2022

- Major GPA: 3.78/4.0

Selected Research & Projects

Research Scientist I, Georgia Tech Research Institute – Atlanta, GA June 2023 – June 2026

- Designed a real-time distributed ingestion platform for radar/EW systems, including ZeroMQ/Protobuf/gRPC telemetry and control services, enabling high-rate IQ/PDW streaming and system-wide coordination across heterogeneous compute nodes under strict latency and bandwidth constraints.
- Built hardware-in-the-loop testbeds integrating FPGA-based jammers, SDRs, and synthetic environments to evaluate scheduling and arbitration policies under realistic contention and timing constraints.
- Characterized latency-throughput and tail-behavior tradeoffs across workload regimes, identifying how resource-allocation decisions under contention produce the tail amplification patterns that motivate my scheduling research.

Awards: Outstanding SW/Algorithm Design/Prototype Award, Innovative Research Award, Distinguished Researcher Award — GTRI, 2024-2025

Senior Software Engineer, Systems & Technology Research – Remote June 2026 – Present

- Investigating cross-process coordination tradeoffs across multiple messaging paradigms (MPI, ZeroMQ, Zyre, TCP) to inform infrastructure design for distributed open-systems radar/sensor pipelines.
- Integrating a search-and-track processing chain into a modular radar development framework, extending the architecture toward end-to-end distributed sensor processing and evaluation.

Research Interests

Real-time and distributed systems; embedded, edge, and cyber-physical systems; scheduling and resource arbitration; tail-latency optimization; performance isolation and policy enforcement under contention; uncertainty-aware and risk-sensitive control of dynamic systems.

Research Publications & Preprints

- **Robert Guy.** *Risk-Aware Guarded Scheduling for Tail Latency Control in Distributed Systems*. Zenodo Preprint, 2026. <https://doi.org/10.5281/zenodo.20030902>
- **Robert Guy.** *Predictive Arbitration for Stateful Distributed Pipelines using Sequence Models*. Zenodo Preprint, 2026. <https://doi.org/10.5281/zenodo.19229146>
- **Robert Guy.** *Adaptive Priority Arbitration for Stateful Distributed Pipelines*. Zenodo Preprint, 2026. <https://doi.org/10.5281/zenodo.19026183>.
- **Robert Guy.** *Distributed Scene Update (DSU): A Middleware Pattern for Real-Time Partial State Synchronization*. Zenodo Preprint, 2026. <https://doi.org/10.5281/zenodo.19010080>